

TFM: “Towards Interpretable Twin Support Vector Machines with Probabilistic Class Membership”

Feature Selection:

- Model agnostic (need to select n or select it using cross-validation as hyperparameter):
 - F-score (Tomar, D., & Agarwal, S. (2014)): n^* (are those n^* alw. the same?)
 - Recursive Feature Elimination (Yang, Z. M., He, J. Y., & Shao, Y. H. (2013)): n^* (only one training sample to do RFE?)
- Embedded:
 - Individual selection per hyperplane (modify such that global selection is $\leq n^* \rightarrow |n_1 \cup n_2| \leq |n^*|$):

- Bai, L., Wang, Z., Shao, Y. H., & Deng, N. Y. (2014)

$$\begin{aligned} \min_{\mathbf{w}_1, b_1, \xi_2} \quad & \|\mathbf{w}_1\|_1 + c_1 \|A\mathbf{w}_1 + \mathbf{e}_1 b_1\|_1 + c_3 \mathbf{e}_2^\top \xi_2 \\ \text{s.t.} \quad & -(B\mathbf{w}_1 + \mathbf{e}_2 b_1) \geq \mathbf{e}_2 - \xi_2, \\ & \xi_2 \geq 0, \end{aligned}$$

and

$$\begin{aligned} \min_{\mathbf{w}_2, b_2, \xi_1} \quad & \|\mathbf{w}_2\|_1 + c_2 \|B\mathbf{w}_2 + \mathbf{e}_2 b_2\|_1 + c_4 \mathbf{e}_1^\top \xi_1 \\ \text{s.t.} \quad & (A\mathbf{w}_2 + \mathbf{e}_1 b_2) \geq \mathbf{e}_1 - \xi_1, \\ & \xi_1 \geq 0. \end{aligned}$$

- Joint selection:

- Maldonado, S., & López, J. (2017).

$$\begin{aligned} \min_{\mathbf{w}_k, b_k, \xi_k} \quad & \frac{1}{2} \left(\|A\mathbf{w}_1 + \mathbf{e}_1 b_1\|^2 + \|B\mathbf{w}_2 + \mathbf{e}_2 b_2\|^2 \right) + \lambda \sum_{j=1}^n \|\mathbf{w}^{(j)}\|_\infty \\ & + \frac{c_1}{2} \left(\|\mathbf{w}_1\|^2 + b_1^2 + \|\mathbf{w}_2\|^2 + b_2^2 \right) + c_2 (\mathbf{e}_1^\top \xi_1 + \mathbf{e}_2^\top \xi_2) \\ \text{s.t.} \quad & A\mathbf{w}_1 + \mathbf{e}_1 b_1 - A\mathbf{w}_2 - \mathbf{e}_1 b_2 \geq \mathbf{e}_1 - \xi_1, \\ & B\mathbf{w}_2 + \mathbf{e}_2 b_2 - B\mathbf{w}_1 - \mathbf{e}_2 b_1 \geq \mathbf{e}_2 - \xi_2, \\ & \xi_1 \geq 0, \xi_2 \geq 0, \end{aligned}$$

- Bai, L., Wang, Z., Shao, Y. H., & Deng, N. Y. (2014)

$$\begin{aligned} \min_{u_1, u_2, \gamma_1, \gamma_2, \eta_1, \eta_2, E} \quad & \sigma e^\top E e + \lambda (\|u_1\|_1 + c_{11} \|K(A_1 E, A E) u_1 + \gamma_1 e\|_1 \\ & + c_{12} \|\eta_1\|_1) + (1 - \lambda) (\|u_2\|_1 + c_{21} \|K(A_2 E, A E) u_2 \\ & + \gamma_2 e\|_1 + c_{22} \|\eta_2\|_1) \\ \text{s.t.} \quad & -(K(A_2 E, A E) u_1 + \gamma_1 e) \geq e - \eta_1, \quad \eta_1 \geq 0, \\ & K(A_1 E, A E) u_2 + \gamma_2 e \geq e - \eta_2, \quad \eta_2 \geq 0, \\ & E = \text{diag}(1 \text{ or } 0), \end{aligned}$$

Ideas: include performance/fairness constraints; use bootstrapped feature importance for stability; question how much current feature selection results depend on the training sample.

Probabilities estimation (first, calculate scores as in *Shao, Yuan-Hai, et al. (2012)*):

- Parametric methods:

- Temperature

$$P(Y = 1|f(x)) = \frac{1}{1 + e^{f(x)/T}}$$

- Platt (as in *Shao, Yuan-Hai, et al. (2012)*)

$$P(Y = 1|f(x)) = \frac{1}{1 + e^{Af(x)+B}}$$

- Polynomial logistic

$$P(Y = 1|f(x)) = \frac{1}{af(x)^2 + bf(x) + c}$$

- Generalized logistic

$$P(Y = 1|f(x)) = \frac{1}{(1 + e^{-k(f(x)-f_0)})^v}$$

- Penalized spline (fixed basis)

$$P(Y = 1|f(x)) = \frac{1}{1 + e^{\sum_{k=1}^K \beta_k B_k(f(x))}}$$

- Bootstrapping with probabilities (similar to *Benítez-Peña, Sandra, et al. (2024)*, but using $P(\cdot)$ instead of $f(\cdot)$)

- Non-parametric methods:

- Isotonic regression

$$P(Y = 1|f(x)) = g(f(x)), \quad \text{where } g(\cdot) \text{ non-decreasing.}$$

- Histogram binning

$$\hat{P}_k = E[Y|f(x) \in \text{bin } k]$$

- Kernel smoothing (Nadaya-Watson)

$$P(Y = 1|f(x)) = \frac{\sum_{i=1}^n K_h(f(x) - f(x_i)) y_i}{\sum_{i=1}^n (f(x) - f(x_i))}$$

- kNN Calibration

$$P(Y = 1|f(x)) = \frac{1}{k} \sum_{i \in N_k(f(x))} y_i, \quad \text{first if } y_i = -1 \rightarrow y_i = 0$$

- CDF-based (quantile calibration, using empiric CDF)

$$P(Y = 1|f(x)) = \hat{F}_{Y=1}(s)$$

- Bootstrapping (as in *Benítez-Peña, Sandra, et al. (2024)*)